



LUVATA PA12 COATED BUSBARS
**Boosting efficiency
in electric vehicles**

Introduction

The blooming electric vehicle (EV) industry has increased the need for more durable and high-precision components. The evolution of large EV battery packs necessitates materials that can withstand higher voltages, ensure thermal conductivity and fit all in confined spaces while ensuring safe, cost-efficient and reliable operation.

Expertise in automotive and EV advancements

Luvata addresses this with PA12 coated busbars designed explicitly for the EV market. Compared to rigid stamped busbars, these busbars offer enhanced thermal and electrical performance and increased reliability. Unlike other manufacturers, Luvata offers busbars not only as cable but also as precision-bending products to fit customers' needs.

As a leading manufacturer of resistance welding electrodes and accessories, Luvata has long-held relationships with nearly every automotive manufacturer globally. Luvata's background in partnering with the automotive industry brings extensive know-how to the EV industry.

The company's products have seen improvements related directly to keeping up—and ahead—with the development of the EV industry, which is also manifested in Luvata's PA12 insulated busbar and its refined characteristics.





Market background of the EV industry

The electric vehicle market has faced multiple changes in recent years, including technological advancements, regulatory changes, and evolving consumer preferences.

As the market is estimated to face robust growth, EVs are expected to account for a significant share of new vehicle sales globally. In 2024, electric car sales are projected to reach around 17 million, accounting for over 20% of all cars sold worldwide. Growth in electric car sales is also picking up in emerging economies.

Over 20 major car manufacturers, representing over 90% of global car sales in 2023, have set electrification targets. By 2030, more than 40 million electric cars could be sold annually if current policy settings are maintained. Committed investments in battery manufacturing are sufficient to meet the projected demand under various scenarios, with global EV battery manufacturing capacity far exceeding current demand.

Ensuring a stable supply of critical materials like lithium, cobalt, and nickel remains a challenge in the market. Companies are investing in securing supply chains and developing alternative materials globally. The industry is also placing a greater emphasis on sustainable practices, including sourcing raw materials for batteries, like copper, and recycling used batteries.

As the market and consumer expectations for vehicles increase, the need for more efficient batteries is evident. Manufacturers that can produce EVs with superior battery performance have a competitive advantage. Effective batteries deliver power more efficiently, enhancing the vehicle's overall performance and supporting the broader adoption of EVs, lowering greenhouse gas emissions and reducing reliance on fossil fuels.

Technical specifications and applications of PA12 coated busbars

The design and construction of busbars and interconnect systems are crucial for EV battery performance. The Luvata high-adhesion PA12 coated copper strip can withstand a voltage of nearly 30kV and is thus commonly used for battery connections.

PA12 coated busbars have been implemented in battery pack interconnects for battery modules, onboard inlet charging to the battery pack, power inverters, DC-to-DC Converts, and auxiliary connectors to heaters and AC compressors in EVs. The key benefits observed in these applications are easy installation in limited space, minimized weight and impedance, and safe and reliable operation in EVs' challenging conditions.

The PA12 coated busbar is 30% to 40% smaller in size when compared to a cable rated at the same capacity. Also, with a flat surface, the heat dissipation is higher. Thus, they have been used to replace cables and ridged busbars in power transmission. Additionally, these busbars are being used in data centers and the renewable energy industry for innovative applications where flexible connections with high power requirements and increased flexibility are needed.

Technical info on PA12 coated busbars

- Withstands voltage to nearly 30kV
- Soft temper (R200): easy to form and bend
- Standard-grade oxygen-free copper (Cu-OF)
- Excellent coating adhesion
- Flat cable: width 8-20mm, thickness 2-6mm, full round edges

Oxygen-free copper in busbars

Polyamide PA12 insulated busbars are made from oxygen-free copper (Cu-OF). Unlike regular copper, oxygen-free copper attains superior conductivity and bendability, provides the highest degree of cold formability and minimizes the risk of edge cracks during severe edge bending applications. Additionally, oxygen-free copper eliminates the potential of oxide inclusions when crack initiations could occur during cold working.

The absence of oxygen in the component allows more currents to pass through the same conductor size over other forms of copper, giving it an advantage for applications in the EV market.



PA12 as an insulation material – the ability to withstand voltages and precision-bending

Electrical insulation is also an essential factor with busbars and for the whole power pack. The challenge is understanding the materials, the high currents and voltages, and the necessity for efficient heat dissipation.

PA12 is ideal for the mentioned situations mainly for its good insulating properties and excellent adhesion with oxygen-free copper. These properties are complemented by excellent durability and bendability, improved wear resistances, low water absorbency, good dimensional stability and high chemical resistance. PA12, as an insulation material, contributes to the busbars' overall performance and safety, bringing with it the

ability to withstand harsh environments and reliable performance in environments where oil and fuels are present. This makes PA12 an excellent insulation material, particularly for the EV industry.

The key advantages of using PA12 insulated busbars in electric vehicle power systems compared to other busbars are numerous. They offer improved flexibility and reduced space requirements, allowing greater misalignment between connection points. This will enable busbars to be designed with unique bends that allow them to expand in directions opposite to the connection points thermally, reducing the forces on bolted and welded connections.

Designing and implementing PA12 insulated busbars

Busbars in EVs are commonly made from thick, flat, rigid copper wire. It is bent across both its width and thickness and sometimes twisted. The PA12 coated copper strip is available in a very soft temper, allowing easy forming and bending. The product can be bent into sharp angular shapes and loops while maintaining tight adhesion of the insulation. It can be joined easily using all welding and brazing methods and is ideal for manufacturing processes that require extreme deformability. By using already bent components, system costs can be lowered as one processing step is eliminated entirely.

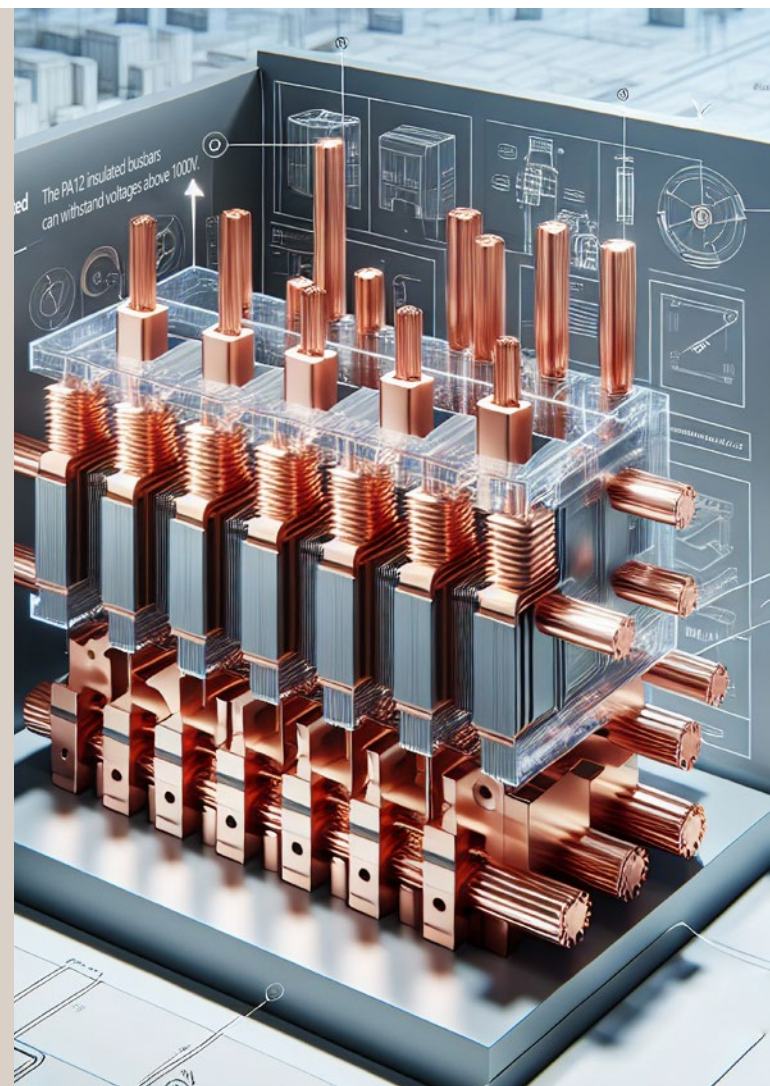
PA12 coated busbars offer better flexibility than their competitors. This reduces any possible rework after assembly, particularly when working with components that are not in perfect alignment. Another

challenge with implementing PA12 insulated busbars in electric vehicle applications is the locations of different connection points. These will determine bending ratios and twist locations to ensure a perfect fit for the component. Regarding safety, the design of the initial constraint for the cross sections is crucial. This ensures the effective carrying of the current without exceeding the safe operating temperatures.

As implementing busbars is often precise, it also requires minimal tolerance. Uniformity in quality and size within a produced batch and particularly beyond that is necessary. This is also one of Luvata's main promises to its customers.

Key benefits of PA12 insulated busbars

1. PA12 coated busbars offer improved flexibility, reduced space requirements, enhanced thermal performance, and increased reliability over ridged stamped busbars.
2. Busbars' high-temperature resistance ensures safe and reliable operation during charging and under high-voltage loads.
3. Busbars help reduce the forces on bolted and welded connections since they can be designed with unique bends that allow them to thermally expand in directions opposite to the connection points.
4. Busbars are more flexible than other stamped busbars, allowing for greater misalignment between connection points.
5. The PA12 insulated busbars can withstand voltages above 1000V.



Enhancing sustainable operations

Luvata's commitment to Environmental, social and corporate governance standards (ESG) sits tight at the center of its strategy and actions.

As a global company with the power to contribute to a better future, the company's vision is to be the leading provider of copper solutions for the benefit of People, Society and the Earth. Luvata's products and solutions are thus building towards carbon neutrality and sustainable development.



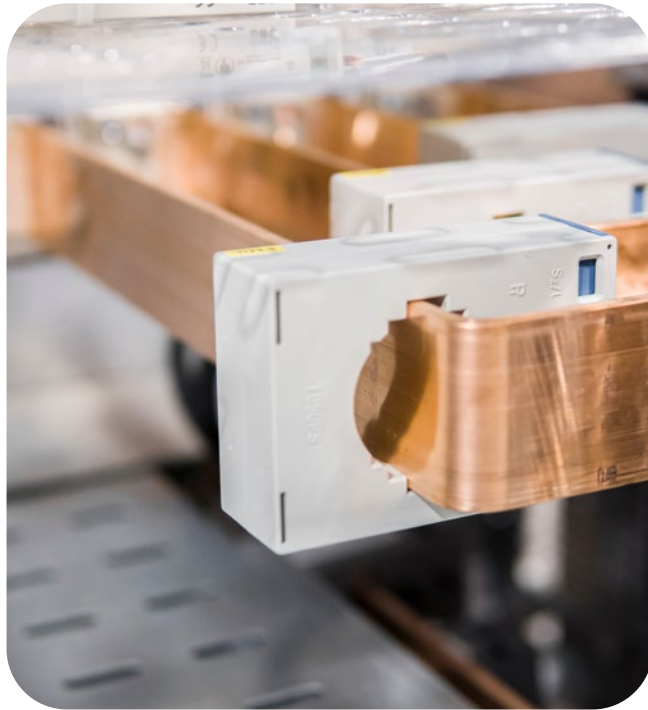
Copper with low-carbon footprint in PA12 insulated busbars

The electric vehicle industry and other industries relying on non-renewable resources are now examining the supply chain and production line with sustainability questions. Luvata's production processes consider tightened sustainability goals, consumer preferences, and growing awareness of ecological questions. With busbars, one of the successful responses to this demand is using copper with a low-carbon footprint.

The production of low-carbon copper involves processes that emit fewer greenhouse gases than traditional methods, reducing the overall carbon footprint of products using this material.

The low-carbon copper used in producing PA12 coated busbars comes from Luvata's long-term supplier, enabling the customers to comply with increasingly stringent environmental regulations and carbon reduction targets.

Moreover, low-carbon copper offers improved performance regarding high conductivity and durability. The copper maintains the excellent electrical conductivity of traditional copper and retains the durability and reliability of standard copper, ensuring long-lasting performance in various applications, like electronics, renewable energy, and EVs.



Safe and stable production chain

Amid geopolitical uncertainties in Europe, the market has worried about the sufficient availability of copper. Luvata uses partners with secure and stable backgrounds to ensure sustainable and ongoing operations. With an extensive history in component manufacturing in the automotive and vehicle industry, Luvata boasts a deep-rooted expertise in copper processing dating back to the 1940s. The company is still working with its old local supply chain in Pori, Finland, processing the raw materials and manufacturing the products independently at its site. The whole production chain is in Luvata's hands, bringing ease of mind to customers as well.

Luvata's steadfast commitment to a safe and uninterrupted production flow underscores its dedication to customer satisfaction and operational excellence. As the supply of raw materials for Luvata is secured, so is the supply of the end product.

Developing the EV industry for future demands

As the EV industry keeps growing and developing its technologies, the requirements for components like busbars will keep becoming more particular and demanding. One of the challenges this may bring forth is managing process variability and reproducibility. Cooperation with companies with long-lasting relationships with customers, suppliers and staff will become essential to development - the know-how and dependability from experience will be vital.

Emerging technologies and advancements mainly enhance the performance and functionality of busbars or utilize existing knowledge in other fields. For example, PA12 coated busbars are currently more commonly used in light vehicles, but the adaptation to large equipment programs is already under development. Another future development will be developing integrated shielding and filtering for busbars and integrating thermal management systems into the busbars. The former will be critical to electrical system design, where EMI and electrical noise must be minimized.

Luvata's consistent experience and commitment to the vehicle industry make it a trustworthy partner for future needs and developments.



Sources:
International Energy Association (IEA), Global EV Outlook 2024, April 2024 | Electrical Engineering Portal, June 2024
E-Mobility Engineering, September/October 2023 | International Copper Association, Sustainable Copper, July 2024



About Luvata

Luvata is a world leader in copper manufacturing and related engineering services. It brings together people, innovation and technology to create solutions in areas such as e-mobility, healthcare, and in electrification & the green transition. The company's success is driven by innovation, lean production and value-added sales, and its vision is to provide copper solutions which benefit people, society and the earth. Employing over 1,500 staff on four continents, Luvata works in partnership with customers such as ABB, CERN and Siemens. Luvata is a group company of Mitsubishi Materials Corporation.

Find out more about Luvata and offerings to the e-mobility and electric vehicle industry:
<https://www.luvata.com/industries/e-mobility>

luvata.com

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PA12 coated copper strip

Bendability for the benefit of electric vehicles

Luvata's metallurgical, manufacturing and insulation expertise continues to support the future of EVs.

Through our continuous improvement, Luvata has succeeded in developing a high-adhesion PA12 coated copper strip.

Luvata PA12 coated copper strip is specifically designed for the growing EV market. It is able to withstand voltage of nearly 30kV and commonly used for battery connections.

Luvata PA12 coated copper strip is offered in a very soft temper, so it is easy to form and bend. The product can be bent into sharp angular shapes and even loops, meanwhile the adhesion of the insulation remains tight.

Luvata standard grade oxygen-free copper (Cu-OF) is a high purity copper that is immune against hydrogen embrittlement. It is used in applications where high electrical and thermal conductivity are essential.

It can be easily joined using all welding and brazing methods and is ideal for manufacturing processes requiring extreme deformability.



About Luvata

Luvata is a world leader in metal solutions manufacturing and related engineering services to industries such as renewable energy, automotive, healthcare, and power generation and distribution. The company's continued success is attributed to its longevity, technological excellence and strategy of building partnerships beyond metals. Employing approximately 1,400 staff in 7 countries, Luvata works in partnership with customers such as ABB, CERN, Siemens and Toyota. Luvata is a group company of Mitsubishi Materials Corporation.

Technical specifications

Alloy	standard grade oxygen-free copper (Cu-OF)
Dimensions	width 8-20mm, thickness 2-6mm, full round edges
Soft temper	R200
Adhesion	excellent coating adhesion
Flammability rating	tested according to UL 94 HB

Packaging

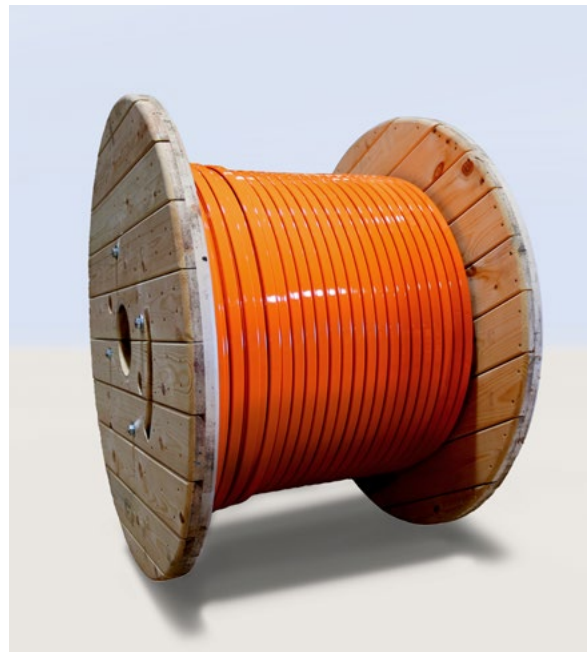
Luvata high-quality PA12 coated copper strip is cut to the exact length of customer specifications and delivered on reels.

Support in overcoming the challenges facing the e-mobility industry

Luvata is focusing on copper's high thermal conductivity in designing next generation products that overcome customers' challenges in terms of charging times, heating, cooling, driving ranges and infrastructure.

Luvata recognizes these challenges as opportunities and has made improvements specifically targeting the accelerated pace of e-mobility research and development by offering:

- Custom designed semi-products
- Insulated semi-products or components ready for assembly
- High performance alloys
- Specifications with extreme tolerances
- Accelerated prototype manufacturing



Luvata PA12 coated copper strip on wooden reel.



Luvata PA12 coated copper strip available in all RAL standard colours.

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